



April 2024

Total Marks:-30

Answer:

8. What is the purpose of TF-IDF in text processing and information retrieval systems?

- A) Image processing
- B) Keyword extraction
- C) Data storage
- D) Encryption

Answer: _____

9. Which of the following correctly defines reinforcement learning in Artificial Intelligence systems?

- A) Learning with labeled data
- B) Learning through rewards and penalties
- C) Learning without any data
- D) Learning using images only

Answer: _____

10. What is the main role of an agent in an Artificial Intelligence system interacting with an environment?

- A) Store data
- B) Make decisions and perform actions
- C) Compile programs
- D) Display results

Answer: _____

11. Which library is widely used for Natural Language Processing tasks in Python programming?

- A) NumPy
- B) NLTK
- C) Pandas
- D) OpenCV

Answer: _____

12. What is the primary function of stopwords removal during text preprocessing in NLP tasks?

- A) Remove important keywords
- B) Remove common words like “is”, “the”
- C) Add extra words
- D) Encrypt text

Answer: _____

13. Which concept in game theory refers to a situation where no player benefits by changing their strategy alone?

- A) Zero-sum game
- B) Nash Equilibrium
- C) Heuristic
- D) Tokenization

Answer: _____

14. What is the function of the open list in the A search algorithm used in pathfinding problems?

- A) Store explored nodes
- B) Store nodes to be explored
- C) Store final path
- D) Store heuristic values

Answer: _____

15. What is the purpose of Optical Character Recognition (OCR) in AI-based systems and applications?

- A) Audio processing
- B) Text extraction from images
- C) Video editing
- D) Networking

Answer: _____

16. In reinforcement learning, what does a reward signal represent when an agent interacts with an environment and takes actions?

- A) Punishment for incorrect actions
- B) Feedback indicating success or failure
- C) Cost of computation
- D) Random output

Answer: _____

17. What is the key difference between supervised learning and reinforcement learning in terms of how feedback is provided to the model?

- A) Supervised learning uses rewards
- B) Reinforcement learning uses labeled data
- C) Reinforcement learning uses interaction-based feedback
- D) No difference

Answer: _____

18. In Natural Language Processing, what is the role of tokenization when preparing raw text data for further processing?

- A) Encrypt text
- B) Split text into smaller units like words
- C) Store data
- D) Compress files

Answer: _____

19. What is the purpose of cosine similarity when comparing two documents in AI-based answer evaluation systems?

- A) Measure distance
- B) Measure similarity between text vectors
- C) Sort documents
- D) Encrypt data

Answer: _____

20. Which technique is used to extract important keywords from a dataset by considering both frequency and uniqueness of terms?

- A) Sorting
- B) TF-IDF
- C) Searching
- D) Hashing

Answer: _____

21. What is the role of the heuristic function in the A search algorithm when finding the shortest path in a graph?

A) Exact cost calculation

B) Estimated cost to goal

C) Random value

D) Final output

Answer: _____

22. In the A search algorithm, what is the significance of combining path cost and heuristic value for node evaluation?

A) Random search

B) Optimal path finding

C) Data storage

D) Encryption

Answer: _____

23. In game theory, what does a zero-sum game indicate about the relationship between players' gains and losses?

A) All players win

B) One player's gain equals another's loss

C) No loss

D) Random outcome

Answer: _____

24. What is Nash Equilibrium in a competitive game scenario where multiple players choose strategies independently?

A) Best mutual strategy

B) Random decision

C) Worst case

D) No solution

Answer: _____

25. In an AI-based answer evaluation system, how is similarity between student answer and model answer typically calculated?

A) Sorting

B) Cosine similarity

C) Searching

D) Encoding

Answer: _____

26. What is overfitting in machine learning models when trained on a limited dataset with high accuracy?

A) Poor performance

B) Model fits training data too well

C) No learning

D) Random output

Answer: _____

27. What is underfitting in machine learning when a model fails to capture patterns from the training data?

A) Perfect model

B) Poor learning

C) High accuracy

D) Complex model

Answer: _____

28. In NLP preprocessing, why is stopwords removal important when analyzing large text datasets?

A) Increase noise

B) Reduce unnecessary common words

C) Add complexity

D) Encrypt data

Answer: _____

